Supplement to "Slowly Rotating Perfect-Fluid Spheres with an Adiabatic Gas Surface"

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S1 External rotational benchmarks

All dimensionless quantities use the nonrotating mass and radius, with $G = c = 1$: $\mathscr{C} = M/R$, $\bar{I} = I/M^3$, and $\bar{Q} = -QM/J^2$; an oblate star has $Q < 0$. The matching solver is tested against the homogeneous-star table of Chandrasekhar and Miller [1] and the $n = 1$ polytropic curves of Yagi and Yunes [2]. These are external model calculations, not universal-relation fits.

Table S1: Homogeneous-star benchmark. Published values are converted from Table I of Ref. [1], using $\bar{I} = 8(I/R_S^3)$ and $\bar{Q} = Q_{\rm table}/2$, with $R_S = 2M$.

$\mathscr{C}$	$\bar{I}_{\rm pub}$	$\bar{I}_{\rm calc}$	Difference (%)	$\bar{Q}_{\rm pub}$	$\bar{Q}_{\rm calc}$	Difference (%)
0.25000000	8.392	8.39506760	+0.03655	3.2625	3.26301980	+0.01593
0.16666667	16.984	16.98235350	−0.00969	7.8350	7.83589182	+0.01138
0.10000000	43.896	43.89661657	+0.00140	18.8650	18.86252015	−0.01315

For this test the background is the exact constant-density Schwarzschild interior. Its surface density is nonzero: no Love number or monopole mass correction is inferred from this test. The percentage agreement refers to the historical numerical table and does not imply agreement within half the last printed digit of every entry.

The polytropic benchmark uses $P = K\varepsilon^2$, as in Ref. [2]. This differs from the thermodynamic rest-mass polytrope $P = K\rho_0^2$, $\varepsilon = \rho_0 + P$, used in the main equal-compactness comparison. With $E = 8\pi\varepsilon$ and a scale choice $\Pi = E^2$, the logarithmic enthalpy is $h = 2\ln(1+E)$ and $c_s^2 = 2E$.

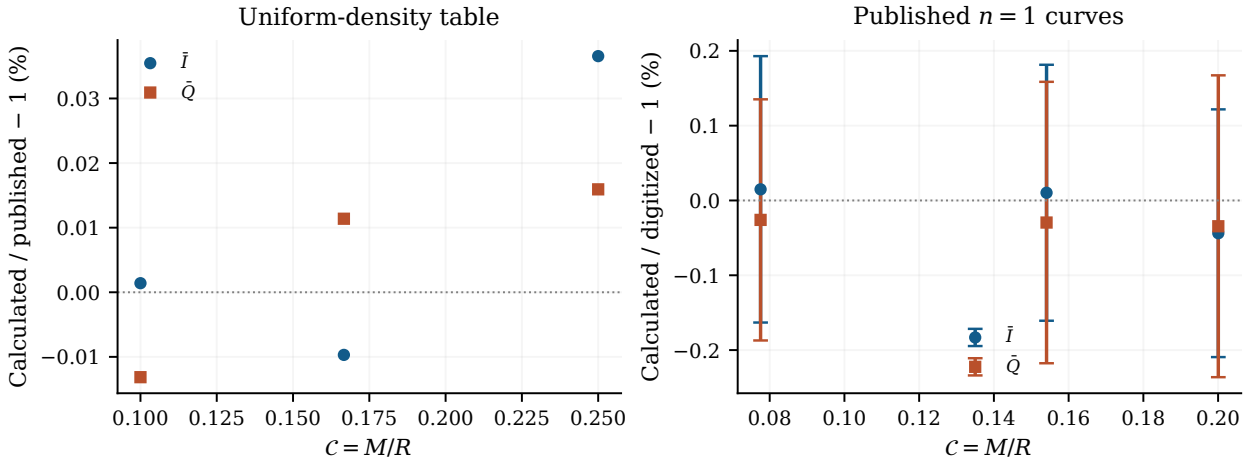


Figure S1: Relative differences from the published homogeneous-star table (left) and digitized $n = 1$ curves (right). The right-hand error bars indicate a conservative graphical-resolution allowance, including both coordinate directions, not statistical errors or uncertainties assigned by the original authors. Plotting code was developed with ChatGPT/Codex assistance.

S2 Vector-figure extraction and thermodynamic limits

The $n = 1$ vertices come from the right-hand panels of Figs. 6 and 7 of Ref. [2], using the original source files `IbarC-PRD.eps` and `QbarC-PRD.eps`. Their $n = 1$ line is orange and solid (`Color11`); the legend text precedes its associated line sample. Let (x_v, y_v) denote the normalized drawing coordinates. The conversions are

$$\mathscr{C} = 0.004\,150^{x_v - 0.15}, \tag{1}$$

$$\bar{I} = 3(1000/3)^{(y_v-0.15)/0.7}, \qquad \bar{Q} = 0.7(300/0.7)^{(y_v-0.15)/0.7}. \tag{2}$$

Ten checked axis ticks agree with these maps within 4.80×10^{-5} in drawing coordinates. The three paired vertices are

$$\begin{aligned}(x_v, y_I, y_Q) = {} & (0.7417, 0.4838, 0.5404),\\ & (0.8787, 0.3318, 0.4036), \quad (0.9308, 0.2782, 0.3384).\end{aligned} \tag{3}$$

Allowing $\pm 10^{-4}$ per coordinate accounts conservatively for rounding and calibration. The solver is evaluated at the central and shifted compactnesses; their response differences are combined with the vertical-coordinate allowance. This gives a graphical allowance of about 0.16–0.21%, compared with maximum observed differences of 0.044% in $\bar{I}$ and 0.035% in $\bar{Q}$. It is not an interval-certified bound on the underlying ODE solutions. The saved vertices and source hashes are included in the reproducibility archive.

Surface evaluation

Write $s = r^2/R^2$, $z = \sqrt{1-s}$ and use $R = 1$. The pressure integral

$$\Pi(z) = \int_0^z t[E(t) + \Pi(t)]g(t)\,\mathrm{d}t, \qquad E + \Pi = -4fa_j, \tag{4}$$

avoids cancellation between order-z^3 terms when the pressure is order z^5. Similarly, $h(z) = \int_0^z t g(t)\,\mathrm{d}t$ and $u_{\mathrm{int}} = \mathrm{expm1}(h) - \Theta$, where $\Theta = P/\rho_0$, resolve the vanishing internal energy. Differentiating $E + \Pi$ directly gives $E_z = (E+\Pi)_z - \Pi_z$ and a stable surface sound speed. Here quantities inside the integrals are composed with $s = 1 - t^2$.

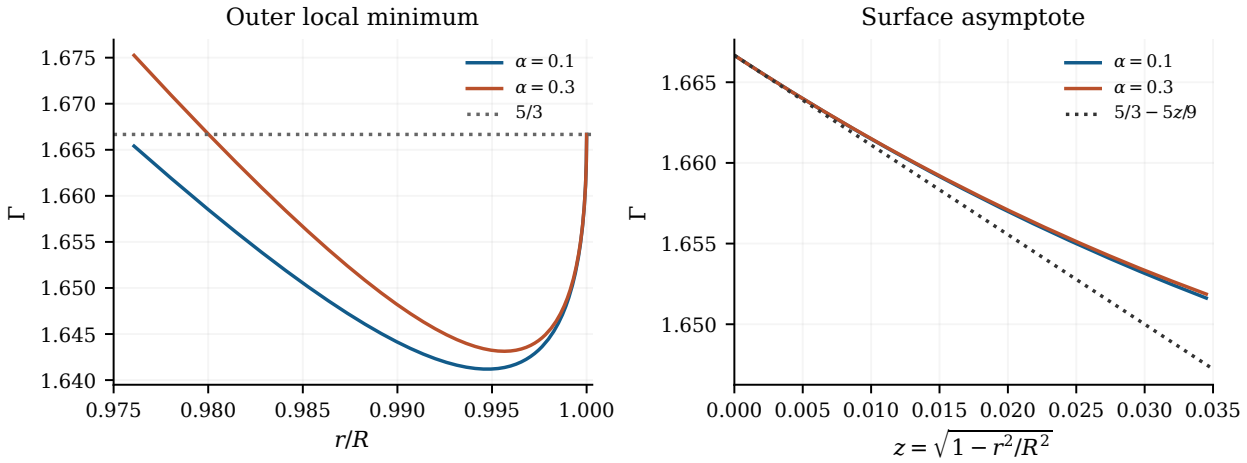


Figure S2: Resolved outer local minima of Γ (left) and comparison with $\Gamma = 5/3 - 5z/9 + O(z^2)$ (right). The limiting value is reached from below. At $\alpha = 0.3$ the outer minimum is local, since the central Γ is smaller. Plotting code was developed with ChatGPT/Codex assistance.

S3 Thermodynamic convergence and pressure per particle

The limiting relations are

$$\Gamma = \frac{5}{3} - \frac{5}{9}z + O(z^2), \qquad \Theta = \frac{g_b}{5}z^2[1 - z/3 + O(z^2)], \qquad \frac{u_{\rm int}}{\Theta} \longrightarrow \frac{3}{2}. \tag{5}$$

Table S2 evaluates these relations at computed interior points. No flattening or smoothing toward an imposed surface value is used.

Table S2: Surface convergence. The limiting entries in the last three columns are $-5/9$, $-1/3$ and $3/2$. Here $g_b = \mathscr{C}/(1-2\mathscr{C})$.

α	z	$(\Gamma - 5/3)/z$	$[5\Theta/(g_b z^2) - 1]/z$	$u_{\rm int}/\Theta$
0.1	10^{-2}	−0.518596	−0.309150	1.50795928
0.1	10^{-3}	−0.551763	−0.330885	1.50082955
0.1	10^{-4}	−0.555175	−0.333088	1.50008330
0.3	10^{-2}	−0.516493	−0.307191	1.50793532
0.3	10^{-3}	−0.551551	−0.330689	1.50082931
0.3	10^{-4}	−0.555154	−0.333069	1.50008329

At six interior radii, an independent five-point derivative check of $\mathrm{d}\ln\rho_0/\mathrm{d}r = (\mathrm{d}\varepsilon/\mathrm{d}r)/(\varepsilon + P)$ has scaled residual below 1.7×10^{-10}. At three inner sample points, algebraic and hydrostatically integrated pressures agree within 6.4×10^{-10} relatively. The metric and hydrostatic logarithmic enthalpies agree within 3×10^{-13} absolutely at the checked points. These are floating-point diagnostics, not validated interval enclosures.

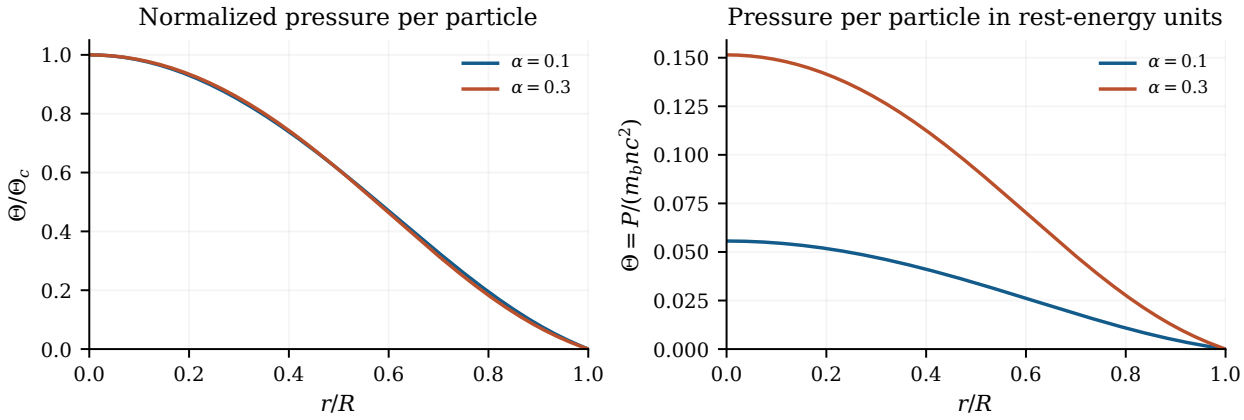


Figure S3: Pressure per particle: normalized Θ/Θ_c (left) and $\Theta = P/(m_b n c^2)$ (right). These curves are not a physical temperature profile. Only for a thermal ideal gas is $k_{\rm B}T = m_b c^2\Theta$; a cold degenerate gas can have the same leading surface law while $T = 0$. The stiff core is outside either free-gas description. Plotting code was developed with ChatGPT/Codex assistance.

References

[1] S. Chandrasekhar and J. C. Miller, *Mon. Not. R. Astron. Soc.* **167**, 63–80 (1974), doi:10.1093/mnras/167.1.63.

[2] K. Yagi and N. Yunes, *Phys. Rev. D* **88**, 023009 (2013), doi:10.1103/PhysRevD.88.023009; original figure sources from arXiv:1303.1528.